

Supporting Information

Different Interfacial Behaviors of Peptides Chemically Immobilized on Surfaces with Different Linker Lengths and via Different Termini

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Simulation Details

We constructed a back-to-back arrangement of two SAMs with an “air gap” to eliminate interactions between the two SAM layers. This was centered in a rectangular volume with periodic boundary conditions to enhance the statistics of calculating behavior of SAM chains and adsorbed water molecules on the SAM surface. To simplify the simulation, the SAM was built by using maleimide terminated molecule with a thiol group reacted with Au. We believe that the results should be applied to the SAM built using silane molecule on silica surface.

The lateral dimensions of the simulation box are 86.63 and 99.94 Å. For Au (111) surface, gold atoms were arranged hexagonally into two layers, assuming the nearest neighbor distance to be 2.884 Å. The positions of the gold atoms were fixed during the simulation. Long chain moieties adsorbed at solid surface have been restricted to physisorbed systems, which hardly achieve dense packing¹. In order to investigate the influence of surface coverage in Mal-PEG2k, two different initial configurations of the system were considered. The system was initially set up with the chains arranged in *all-trans* configurations and standing normal to the surface and it was placed at $c(2\times 1)$ and $c(4\times 2)$ unit cells. Each SAM system included 200 (100 for each side) and 100 (50 for each side) SAM chains for $c(2\times 1)$ and $c(4\times 2)$ packing, respectively. Here, the interaction between Au and S was not explicitly treated. The sulfur atoms in the SAM chains were placed at the 3-fold adsorption sites² and were held using harmonic restraint potential during the simulation. For the comparison, the same system had been made for Mal-EG4 SAM. Each system was fully solvated and about 20 Å thickness of water layer for each side was represented explicitly by using the three-site TIP3 water model³.

All calculations were performed in the NPT (constant particle number, pressure, and temperature) ensemble using CHARMM program⁴. The constant temperature (300 K) and pressure (1 atm along the Z-direction) were maintained by the Nosé-Hoover method⁵ and the Langevin-piston algorithm⁶, respectively. DOMDEC (domain decomposition) module in CHARMM has been used for parallel computing. Parameters and topologies of Mal-PEG2k and Mal-EG4 were generated using the MATCH toolset⁷ with the CHARMM general force field⁸. The van der Waals interactions were smoothly switched off at 10-12 Å by a force-switching function⁹ and the electrostatic interaction were calculated using the particle-mesh Ewald (PME) method¹⁰ with a mesh size of ~ 1 Å for fast Fourier transformation, $\kappa = 0.34 \text{ Å}^{-1}$, and a six-order B-spline interpolation. A time-step of 2 fs was used with the fast SHAKE algorithm¹¹. Equilibration runs of over 5 ns were followed by production runs of over 20 ns with configurations saved every 2 ps for analysis.

References

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